Experiment 3: Information Theory

Aim

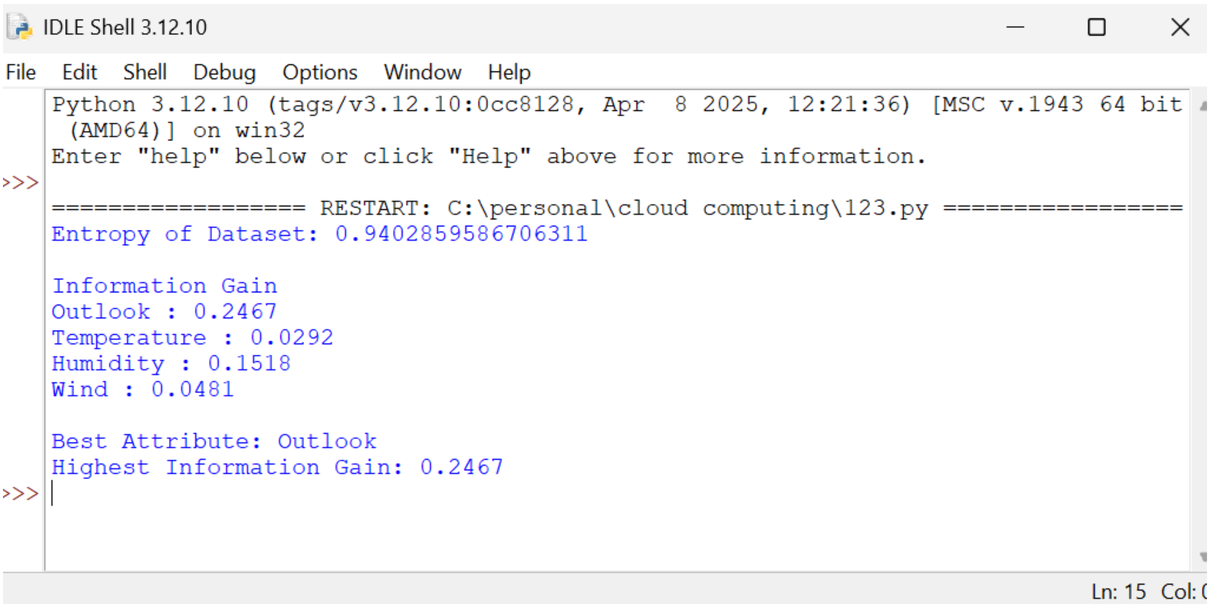
To calculate the entropy and information gain of all input attributes in the Play Tennis Dataset and identify the attribute with the highest information gain.

Program

```
import pandas as pd

from math import log2

data = {

    'Outlook':['Sunny','Sunny','Overcast','Rain','Rain','Rain','Overcast',

               'Sunny','Sunny','Rain','Sunny','Overcast','Overcast','Rain'],

    'Temperature':['Hot','Hot','Hot','Mild','Cool','Cool','Cool',

                   'Mild','Cool','Mild','Mild','Mild','Hot','Mild'],

    'Humidity':['High','High','High','High','Normal','Normal','Normal',

                'High','Normal','Normal','Normal','High','Normal','High'],

    'Wind':['Weak','Strong','Weak','Weak','Weak','Strong','Strong',

            'Weak','Weak','Weak','Strong','Strong','Weak','Strong'],

    'Play':['No','No','Yes','Yes','Yes','No','Yes',

            'No','Yes','Yes','Yes','Yes','Yes','No']

}

df = pd.DataFrame(data)

def entropy(target):

    values = target.value_counts(normalize=True)

    return -sum(p * log2(p) for p in values)

def information_gain(df, attribute, target='Play'):

    total_entropy = entropy(df[target])

    weighted_entropy = 0

    for value in df[attribute].unique():
```

```
            subset = df[df[attribute] == value]
            weighted_entropy += (len(subset)/len(df)) * entropy(subset[target])
        return total_entropy - weighted_entropy
print("Entropy of Dataset:", entropy(df['Play']))
print("\nInformation Gain")
best_attribute = None
best_gain = -1
for col in df.columns[:-1]:
    gain = information_gain(df, col)
    print(col, ":", round(gain,4))
    if gain > best_gain:
        best_gain = gain
        best_attribute = col
print("\nBest Attribute:", best_attribute)
print("Highest Information Gain:", round(best_gain,4))
```

Output



Result

The entropy and information gain were calculated successfully. Among all the attributes, Outlook had the highest information gain and was selected as the best attribute for splitting the dataset.